PROJECT CONTINUITY & ADVANCED DATA ANALYSIS

Sample Superstore | 2014–2017 | Microsoft Power BI

1.PROJECT CONTINUITY FROM WEEK 2

This Week 3 project continues the Week 2 Superstore analysis using the same dataset and Power BI project. Week 2 established approximately $2.0 million in sales, approximately $286,000 in profit and an overall profit margin of about 12.5%. Sales increased from approximately $484,000 in 2014 to approximately $733,000 in 2017, while profit increased from approximately $50,000 to approximately $93,000. Week 3 therefore deepens the analysis rather than restarting it.

The purpose of Week 3 is to investigate monthly sales and profit behaviour, product and subcategory profitability, discount effectiveness, geographic profit concentration and recurring loss-making areas.

2.ADVANCED TIME AND SEASONALITY ANALYSIS

The four-year monthly analysis confirms a strong seasonal pattern. Overall sales were highest in November at $352,461, followed by December at $325,294 and September at $307,650. February was the weakest reported overall sales month at $59,751.

Sales and profit do not move proportionally. Overall February generated $59,751 in sales and $10,295 in profit, while January generated higher sales of $94,925 but slightly lower profit of $9,134. In March 2014, sales were $55,691 but profit only $499, approximately 0.9% margin. July 2014 generated a loss of $841 and January 2015 generated a loss of $3,281. In contrast, October 2016 generated $59,688 in sales and $16,243 in profit, approximately 27.2% margin. November 2016 generated $79,412 in sales but only $4,011 profit, approximately 5.1% margin. In 2017, March achieved approximately 25.1% margin, while November generated the highest listed sales of $118,448 but an approximate margin of only 8.2%.

The key finding is that revenue trends must be interpreted alongside profit and margin. Sales growth alone is not sufficient evidence of stronger performance.

3.DISCOUNT VERSUS PROFIT ANALYSIS

The scatter plot comparing average discount and total profit provides one of the strongest Week 3 findings. The evidence does not support the claim that high discounts automatically cause losses.

Bookcases had an average discount of approximately 21% and total profit of negative $3,473. Supplies had a relatively low average discount of 7.7% but generated negative profit of $1,189. Binders had the highest reported average discount at 37.2% but generated strong total profit of $30,222. Copiers had an average discount of
16.2% and generated total profit of $55,618.

Discount effectiveness is therefore product-specific. Binders demonstrate that high discounting can coexist with strong profitability, while Supplies demonstrate that low discounting does not guarantee profit.

Year-level evidence strengthens this finding. Bookcases were negative in 2014 at negative $346, in 2015 at negative $2,755 and in 2017 at negative $584. Supplies were negative in 2016 at negative $699 and in 2017 at negative $955. Machines became a concern in 2017 with negative profit of $2,869 and an average discount of 30%. In contrast, Binders were profitable in 2015 with $7,597 profit at 38% discount and in 2017 with $7,670 profit at 37.8% discount. Copiers generated $25,032 profit in 2017 at 15.5% discount.

4.PRODUCT AND SUBCATEGORY ANALYSIS

The three most sold subcategories overall are Phones, Chairs and Storage. The three least sold are Fasteners, Labels and Envelopes. The three most profitable subcategories are Copiers, Phones and Accessories. The weakest three by total profit are Tables, Bookcases and Supplies.

Profit margin analysis provides another layer of insight. Labels, Paper and Envelopes have the strongest profit margins, while Tables, Bookcases and Supplies have the weakest margins. High sales volume and high margins are therefore not necessarily the same thing. Labels and Envelopes are among the least sold but among the strongest by margin, creating a potential opportunity for selective growth.

The top three products by total profit are the Canon imageCLASS 2200 Advanced Copier, the Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind, and the Hewlett Packard LaserJet 3310 Copier. The bottom three reported products by profit are the Cubify CubeX 3D Printer Double Head Print, the Lexmark MX611 series Monochrome Laser Printer, and the Cubify CubeX 3D Printer Triple Head Print.

5.GEOGRAPHIC PROFITABILITY ANALYSIS

The three leading states by profit are California at $76,381, New York at $74,039 and Washington at $33,403. California and New York contributed $150,420 combined profit from the reported figures, demonstrating strong concentration in key markets. The three lowest reported states are Wyoming at $100, West Virginia at $186 and North Dakota at $230. These states remain positive but contribute very little and should be reviewed for market potential and investment efficiency.

6.MAIN PROBLEMS AND OPPORTUNITIES

Problem 1: Persistent loss-making areas. Tables, Bookcases and Supplies repeatedly appear among the weakest profitability areas, while Bookcases and Supplies show recurring negative results across multiple years.

Problem 2: Uneven discount effectiveness. Discounts cannot be evaluated uniformly. Binders remain profitable at very high discounts, while Supplies lose money despite low discounts.

Opportunity 3: Seasonal and geographic optimization. Sales are strongest toward the final part of the year, particularly September to December, while profitability is concentrated in major states such as California and New York.

7.OVERALL CONCLUSION

Week 3 confirms the positive growth story identified in Week 2 but reveals a more complex profitability picture. Strong sales do not always translate into strong margins, discount outcomes differ substantially across subcategories, and a small number of products and markets contribute disproportionately to performance. The strongest management priority is product-level profitability monitoring, correction of recurring loss-making areas and protection of high-value products, markets and seasonal opportunities.